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AyuLink - Offline-First Emergency Healthcare Infrastructure for Rural India

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Introduction

Rural India's elderly stay invisible to healthcare until it is too late. A fall at 3 AM can go unnoticed for hours: most monitors need a smartphone, internet and someone nearby.

AyuLink closes that gap: a low-cost wristband, a long-range LoRa link and AI triage alert the doctor, family and ASHA worker in about 3 seconds, with no WiFi at the patient's home.

47 min
avg. rural emergency response (urban: 8 min)

684,000
fatal falls a year worldwide (WHO [3])

Research Objectives

- Detect falls, SOS and tremor (FIDS: rapid shaking) on a ₹2,500 wearable; stream HR, SpO_2 and temperature.
- Deliver alerts over LoRa 433 MHz with no WiFi, 4G or phone at home, and AI-triage each one against JNC 8 / AHA-based limits.
- One platform for doctors, families, ASHA workers and paramedics, in Telugu / Hindi / English.

What's New

- **No WiFi, SIM or smartphone** at the patient's home: the alert rides on LoRa.
- **Offline-first AI triage:** a local LLM at the PHC, with the cloud model only as fallback.
- **Telugu voice alerts** for families, plus an ASHA visit workflow, not just a dashboard.

Methodology

1. **Sense:** ESP32 wristband. MPU6050 flags falls and tremor (FIDS); SOS button; MAX30102 for HR / SpO_2.
2. **Transmit:** LoRa SX1278 433 MHz packet every 2 s (0.8 s in an emergency) to the village gateway.
3. **Process:** FastAPI checks SOS / Fall / FIDS first, then 7 alert types at warning and critical limits.
4. **Triage:** local LLM (Ollama) with Groq LLaMA 3.3 70B fallback gives urgency, cause, action and a 0–100 risk score.
5. **Respond:** red banner, Call 108, Telegram Telugu voice alert, OLED and buzzer on the device.

System Architecture

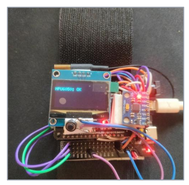
1. **Wristband · ESP32 + sensors**
   fall · tremor · SOS · HR · SpO_2
2. **LoRa 433 MHz · no WiFi / 4G**
   3–5 km rated range · packet every 2 s
3. **PHC Gateway · ESP32 + OLED**
   LoRa → WebSocket bridge
4. **FastAPI + AI Triage Engine**
   thresholds · local LLM + Groq fallback
5. **Doctor · Family · ASHA · EMS**
   dashboards · Telegram · Call 108

Clinical Alert Limits

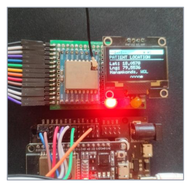
Vital	Warning	Critical
Heart rate (bpm)	≥ 100 or ≤ 55	≥ 120 or ≤ 40
SpO_2	< 94 %	< 90 %
Temp (°C)	≥ 38.0	≥ 39.0 or < 35.0
BP (mmHg)*	≥ 140 / 90	≥ 180 / 120

SOS / Fall / FIDS alert instantly. *BP is entered manually by ASHA / doctor.

Working Prototype



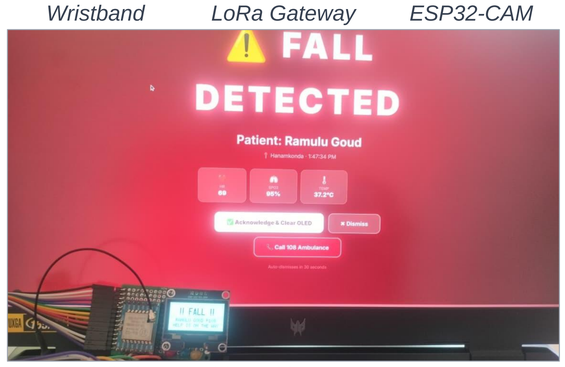
Wristband



LoRa Gateway



ESP32-CAM



Live fall alert: dashboard banner + gateway OLED

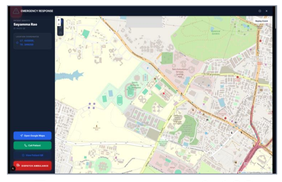
Dashboards & Alerts



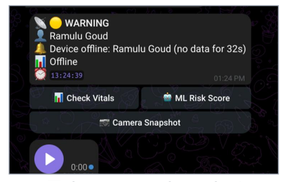
Doctor dashboard



AI triage agent



Emergency map & 108



Telegram voice alert

Prototype Status

Working on hardware: fall (> 1.8 g), tremor and SOS → LoRa → gateway → dashboards, Telegram and AI triage in about 3.2 s.

Simulated in demo: HR / SpO_2 / temp values (MAX30102 calibration pending); AI risk model uses synthetic data.

Next: timed trials, a PHC field pilot, 865–867 MHz licence-exempt band, DPDP-compliant consent.

Results

3.2 s
fall → doctor's screen alert (prototype)

4 portals
doctor · family · ASHA · paramedic

3–5 km
LoRa rated range, no WiFi at home

₹18
per patient / month (est.)

AyuLink vs Typical

	AyuLink	Typical
Connectivity	**Not needed**	WiFi + phone
Range	**3–5 km**	~10 m (BLE)
Cost / month	**₹18**	₹500+
Hardware	**₹2,500**	₹50,000+
Alert time	**< 3 s**	Minutes
AI + language	**Yes · Telugu voice**	No · English only

Typical = BLE / WiFi, smartphone-based monitors (project estimates).

Conclusion

AyuLink is a working prototype that alerts caregivers to a rural fall or SOS in about 3 seconds, with no WiFi or smartphone at home.

Future scope: multi-village LoRa mesh, edge AI, ABHA integration, solar nodes.

References

1. A. Augustin et al., "A study of LoRa: Long range & low power networks for the Internet of Things," Sensors, vol. 16, no. 9, 2016.
2. P. A. James et al., "2014 evidence-based guideline for the management of high blood pressure in adults (JNC 8)," JAMA, vol. 311, no. 5, 2014.
3. World Health Organization, "Falls – Fact sheet," 2021.

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